When Does Hamiltonian Structure Help Control?

A reproducible study of coverage, coordinate priors, model error and policy optimization

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Abstract

Hamiltonian structure is a useful inductive bias, but its contribution to downstream control can be obscured by known mechanics, coordinates, data coverage and policy optimization. We present a reproducible study that separates these factors using controlled pendulum systems, exact Fourier force regressors, neural world models, finite-budget soft actor-critic (SAC), fixed feedback and model predictive control. The central contribution is an experimental decomposition of apparent structure advantages, with independently checked mathematical diagnostics and a retained negative acquisition result. In a validation-selected coupled-pendulum benchmark, Hamiltonian force estimation improves local extrapolation, while global coverage substantially narrows the gap. An incorrectly centered parity prior has an explicit approximation floor, removed by privileged recentering. Independent coupled Duffing validation reproduces the dominant coverage effect. A frozen task-weighted acquisition candidate fails all four primary tracking cells despite reducing its posterior uncertainty criterion. Finally, exact-force control exhibits first-order action-hold error, separating a numerical controller floor from model uncertainty. These findings support conditional, task-specific interpretations of structure rather than a universal Hamiltonian advantage. The study supplies original computational results and a self-contained analytical synthesis; it does not introduce a new general learning algorithm or assert novelty of classical regression, filtering or invariant-design results.

Keywords: Hamiltonian learning; model-based reinforcement learning; experimental design; data coverage; coordinate priors; sampled control.

1 Introduction

A dynamics model can conserve its own learned energy and still predict the wrong physical trajectory. A more accurate dynamics model can also fail to improve a learned policy when optimization is the limiting factor. Both distinctions matter when Hamiltonian world models are used for reinforcement learning.

Our question is: **when does Hamiltonian structure improve cross-energy prediction and control after accounting for coordinate representations, mechanical priors, coverage, regularization and optimization?** We investigate the question through staged, paired interventions. The initial single-pendulum experiments motivate the stronger multidimensional comparisons, rather than supply a stand-alone claim of algorithmic superiority. The main analysis uses finite-dimensional force spaces whose regression objectives can be solved and verified directly. This makes it possible to distinguish model fitting from policy optimization.

The study contributes three linked pieces of original experimental evidence. First, a coverage and coordinate-origin benchmark compares equal-dimensional parity and Hamiltonian subspaces, supplemented by a correctly recentered parity diagnostic. Second, a predeclared acquisition screen demonstrates a gap between a task-weighted covariance objective and actual held-action tracking. Third, an independent quartic-system validation and action-hold refinement check test the main alternative explanations. Classical mathematical identities connect these experiments; they are attributed as background rather than promoted as new theorems.

This is an original research manuscript reporting newly executed simulations and analyses. It is not a certificate that no similar empirical observation has appeared previously, and it is not a claim that the failed candidate is a novel successful algorithm. The manuscript is suitable as a reproducible research report and a basis for scholarly review. Submission, venue fit and acceptance require independent assessment.

2 Related work and scope of originality

Hamiltonian neural networks parameterize a scalar energy and obtain the vector field through canonical differentiation [1]. Controlled Hamiltonian learning has also been developed in Symplectic ODE-Net [2]. These works motivate the structural model families; our convex feature experiments do not replicate their full architectures.

Gruver et al. already analyze second-order modeling and coordinate complexity as explanations of Hamiltonian-network performance [3]. Accordingly, deconstructing physical priors is not a new general agenda. Our contribution is the particular coverage intervention, coordinate-origin comparison, controlled optimization separation and unsuccessful acquisition record, together with independently reproducible diagnostics. Claims about the source of performance are confined to the systems and model classes tested here.

Hamiltonian and odd-symmetry kernel/random-feature models are established approaches [4]. Our Fourier heads are transparent finite-dimensional comparators, not exact implementations of that published kernel algorithm. Task-optimal exploration in linear and nonlinear dynamics already uses task-dependent estimation metrics [5,6]. Bayesian active reinforcement learning [7] and transition-occupancy matching [8] address additional policy-relevant information. The spectral acquisition used here specializes weighted experimental design to periodic tracking; neither its weighted trace nor Gaussian covariance update is new. We do not compare against full implementations of those algorithms and cannot claim to outperform them.

Invariant-design symmetrization is classical [9,10]. The torus specialization below explains why frequency weighting alone need not favor nonuniform query locations. It is explanatory analysis added after the frozen gate, with assumptions that intentionally exclude the actual fixed local pilot.

3 Systems, representations and information access

3.1 Controlled mechanics

We study separable systems with known unit kinetic energy and torque ports,

$$H(q,p) = \frac{1}{2}\|p\|^2 + V(q), \qquad \dot{q} = p, \quad \dot{p} = F(q) + u, \quad F = -\nabla V.$$

Every multidimensional comparator knows these kinematics and input channels. Thus the measured structural effect concerns the force representation, not discovery of unknown inertia or canonical coordinates. With control, the true energy balance is $\dot{H} = p^\top u$. Energy diagnostics use signed work and do not imply energy conservation during actuation.

The periodic potential is

$$\begin{aligned} V_s(q) = 1 &- \cos(q_1 - s_1) + 0.8[1 - \cos(q_2 - s_2)] \\ &+ 0.3[1 - \cos(q_1 - q_2 - s_1 + s_2)] \\ &+ \varepsilon[1 - \cos(3(q_1 - s_1) - (q_2 - s_2))]. \end{aligned}$$

Canonical, misspecified and shifted conditions use $(\varepsilon, s) = (0,0)$, $(0.06, 0)$ and $(0.06, (0.45, -0.35))$. The shifted condition is a coordinate-origin probe of the same physics, not independent physical replication.

The independent validation uses a coupled Duffing potential on $\mathbb{R}^2$,

$$V_D(x,y) = 0.5x^2 + 0.65y^2 + 0.1(x-y)^2 + 0.25x^4 + 0.2y^4.$$

This changes domain and functional family; its validation concerns force estimation only. No Duffing SAC or control-transfer result is claimed.

3.2 Force spaces

The periodic force expansion uses twelve representatives,

$$\mathcal{K} = \{(a, b) : a = 1, 2,\ b = -2, \ldots, 2\} \cup \{(0, 1), (0, 2)\},$$

$$F(q) = c + \sum_{k \in \mathcal{K}} [A_k \cos(k^\top q) + B_k \sin(k^\top q)].$$

Unrestricted force has 50 coefficients. Odd force sets $c = A_k = 0$ and has 24. Hamiltonian force has zero mean and coefficients parallel to k, also with 24. Odd Hamiltonian force has 12. Equal parameter count removes one confound; it does not make the two 24-dimensional spaces identical in geometry or expressivity. Soft mean/curl penalties retain 50 coefficients, with statistical effective degrees of freedom depending on regularization.

The Duffing comparison uses all force monomials of degree at most three (20 coefficients), odd-degree force monomials (12), or gradients of nonconstant potentials of degree at most four (14). Dimensions are explicitly different. No Duffing result is presented as a parameter-matched structure comparison.

3.3 Budgets and split discipline

The main coverage experiment uses 4,096 training and 1,024 independent validation force pairs per condition, three world seeds and local/global coverage fractions 0, approximately 0.1 and 1. Sample counts are fixed. Ridge and soft penalties are chosen by validation force error before fresh test evaluation. There are 162 selected models from 1,620 candidates. Selection is not based on policy return.

The final diagnostic protocol is separately frozen before its outcomes, after the earlier results are known. It uses five fresh seeds, 512 Duffing labels per coverage condition, noise standard deviation 0.03 and fixed ridge 10^{-6}. Local configurations lie in $[-0.35, 0.35]^2$ and global/test configurations in $[-1.5, 1.5]^2$. A shared 4,096-point test estimates global force RMS. Prefix replacement and shared noise pair coverage conditions. This diagnostic tests a specified explanation rather than supplies an independent preregistration of the whole study.

Generative force queries are synthetic. Reachability, sensor cost, contacts and unknown input dynamics are not modeled. Zero additional truth queries during model-only policy training must not be confused with zero world-model pretraining labels.

4 Analytical decomposition

4.1 Learned energy and identifiable physics

Antisymmetry gives $\nabla \widehat{H}^\top J \nabla \widehat{H} = 0$: exact Hamiltonian flow conserves its own energy. This places no quantitative bound on conservation of the true H without a relation between H and $\widehat{H}$. A smooth potential perturbation supported outside observed configurations can leave all local force labels unchanged and alter unseen dynamics. Structural consistency therefore does not identify the unobserved potential. These are classical facts, illustrated here by explicit work and missing-mode experiments.

On a torus, curl-free force additionally requires zero cycle periods to be a gradient of a periodic potential. A constant force bias has zero curl but nonzero cycle work. Our earlier exact-error intervention holds global force RMS at 0.04 across three error types: conservative perturbations produce bounded true-energy error in the stated construction, while a constant bias produces substantial secular work. This is an analytic intervention, not evidence that learned models necessarily exhibit those exact errors.

4.2 Coordinates and approximation floors

Translation rotates each sine/cosine coefficient pair. Full Fourier and Hamiltonian spaces are translation-closed; sine-only odd spaces generally are not. The omitted harmonic $k_* = (3, -1)$ gives the exact squared force approximation floor

$$\inf_{\widehat{F}} \|\widehat{F} - F\|^2_{L^2(\mathbb{T}^2)} = \frac{\varepsilon^2 \|k_*\|^2}{2} = 5\varepsilon^2.$$

For $\varepsilon = 0.06$, the RMS floor is 0.134164. Oddness about an incorrect origin adds

$$\frac{1}{2} \sum_{j \in \{1,2,d\}} a_j^2 \|k_j\|^2 \sin^2(k_j^\top s)$$

to that squared floor, giving approximately 0.4435 for the stated shift. Known-center recentering removes the added parity term. It does not identify an unknown symmetry center for free and does not remove the missing mode.

4.3 Conditional estimation variance

Let $v = T\theta$ encode a structural subspace, D the empirical feature Gram matrix and R the population force-norm metric. For a fixed design and fixed quadratic regularizer, write $A_T = T^\top (D + \lambda R + Q)T$. Independent componentwise noise variance σ^2 gives

$$\mathrm{Cov}(\widehat{\theta} \mid q) = \frac{\sigma^2}{N} A_T^{-1} D_T A_T^{-1}, \qquad D_T = T^\top D T,$$

$$\mathcal{V} = \frac{\sigma^2}{N} \operatorname{tr}(R_T A_T^{-1} D_T A_T^{-1}).$$

After population whitening without soft penalties, this becomes

$$\mathcal{V} = \frac{\sigma^2}{N} \sum_j \frac{d_j}{(d_j + \lambda)^2}.$$

Local coverage can create small design eigenvalues and large noise amplification. A valid restricted subspace changes both dimension and restricted spectrum; broad coverage raises poorly observed directions. This is a variance identity, not total risk, a bound on extrapolation bias or the unconditional risk of validation-selected ridge. A separate 2,000-perturbation audit verifies the fixed-ridge calculation without changing selected models.

4.4 Periodic tracking and acquisition

For continuous unsaturated feedback $u = -F_0(q) - \kappa(p - p_*)$, a small force perturbation δF induces first-order error

$$\dot{e}_1 + \kappa e_1 = \delta F(q_0 + p_* t).$$

For mode frequency $\omega_k = k^\top p_*$, uniform-phase steady response has squared contribution

$$\frac{\|A_k\|^2 + \|B_k\|^2}{2(\kappa^2 + \omega_k^2)}.$$

This is a standard stable-filter calculation, local in perturbation amplitude and time. It is not a guarantee for actuator saturation, finite-horizon phase drift, full policy return or optimal RL.

Force-whitened features $B(q)$ use scalar prior precision $\rho = 0.05$ and noise variance σ^2. With $A = \rho I + \sum B_i^\top B_i$, posterior covariance is $\Sigma = \sigma^2 A^{-1}$. Task design minimizes $\operatorname{tr}(W\Sigma)$; Woodbury gives the exact rank-two covariance gain. This covariance excludes misspecification bias.

Proposition (classical invariant-design specialization). Suppose the force space is translation-closed, W commutes with its orthogonal translation representation, and the only fixed information is scalar ρI. For approximate design ξ, $M(\xi) = \int B^\top B \, d\xi$, Haar-uniform design minimizes $\mathrm{tr}[W(\rho I + nM)^{-1}]$.

Proof. Translation maps M to $O_s^\top M O_s$ without changing the criterion. Haar-averaging the translated design gives uniform configuration measure and, by force whitening, information I. Convexity of inverse weighted trace and Jensen's inequality yield $\Psi(I) \le \int \Psi(O_s^\top M O_s)\, ds = \Psi(M)$. This establishes optimality, not uniqueness. The actual nonuniform pilot violates the assumption; finite random sampling is not identical to approximate uniform design. The sine-only class also lacks translation closure. □

4.5 Sample-and-hold error with an exact force model

A true-force controller can have nonzero error because actions are held. The following consistency statement specializes ordinary sampled-control analysis, rather than proposing a new control theorem.

Proposition (finite-horizon first-order action-hold response). Assume F is C^2 with bounded derivatives on a neighborhood of the finite-horizon trajectories, no saturation, $p(0) = p_*$ and exact continuous integration between control updates. At times $t_n = nh$, hold $u_n = -F(q_n) - \kappa(p_n - p_*)$. As $h \to 0$, uniformly at grid times on fixed $[0, T]$,

$$p_n - p_* = h\eta(t_n) + O(h^2), \qquad \dot\eta = -\kappa\eta + \frac{1}{2} DF(q_0 + p_* t)p_*, \quad \eta(0) = 0.$$

Proof. Taylor expansion of the exact held-input flow gives $e_{n+1} = (1 - \kappa h)e_n + \frac{1}{2}h^2 DF(q_n)(p_* + e_n) + O(h^3)$, with uniform remainder on the stated neighborhood. A discrete Gronwall estimate gives $e_n = O(h)$ and $q_n = q_0 + p_* t_n + O(h)$. Substitution and division by h show that e_n/h obeys forward Euler for the displayed linear ODE with local defect $O(h^2)$. Finite-horizon first-order convergence gives $e_n/h = \eta(t_n) + O(h)$, proving the result. The statement is not uniform in arbitrarily long time and excludes clipping. □

This explains an $O(h)$ error scale without claiming that every Phase 7 discrepancy has this cause. The final refinement diagnostic uses the actual unsaturated held controller and a much smaller RK4 integration substep. It is distinct from the clipped primary experiment.

5 Experiments and results

5.1 Coverage and coordinate intervention

For the canonical periodic plant, mean global force RMS changes from 0.344 for unrestricted force and 0.117 for Hamiltonian force under local coverage to 0.0032 and 0.0024 under global coverage. On the misspecified plant, corresponding local values are approximately 0.838 and 0.604, while global values approach the common 0.134 approximation floor. Thus valid structural restrictions help local estimation in this experiment, but do not identify the missing harmonic.

The final recentering diagnostic uses five fresh seeds and fixed ridge, not the earlier validation-selected fits. Under global coverage, the zero-centered odd force has mean RMS 0.44018; the privileged correctly centered model has 0.13399, near the common floor. Local values are 3.08673 and 0.97780. The large local errors show that recentering alone does not solve extrapolation. These results support a coordinate-dependent prior explanation, not an intrinsic inferiority of correctly centered parity.

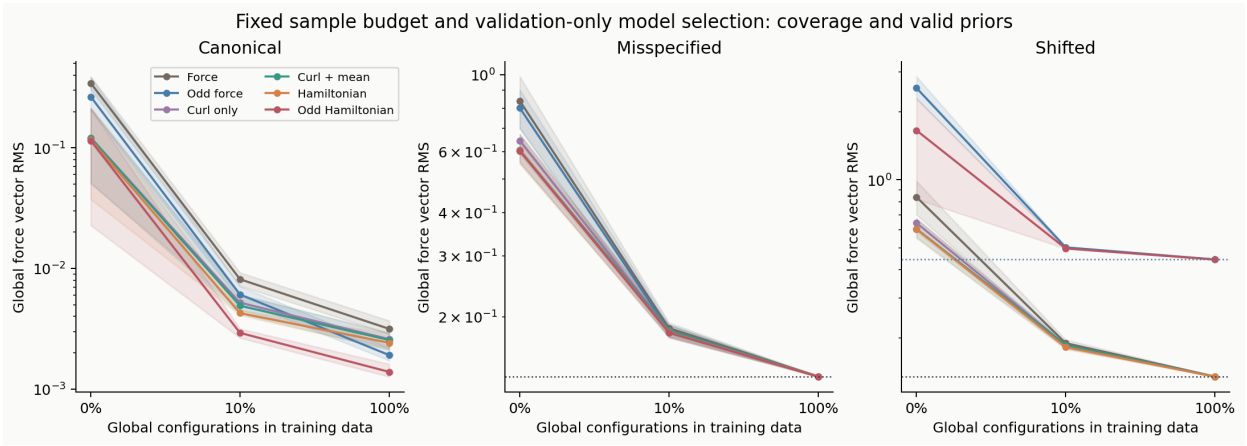


Figure 1: Validation-selected coverage and structural priors.

5.2 Control and RL optimization

The coupled benchmark includes 220 actual feedback/MPC configurations and 57 new SAC training runs. SAC uses final checkpoints, 16,000 transitions and 7,492 updates per run, with three world and three policy seeds in the selected comparisons. All 36 speed-reset interventions are retained. A reset changes the training distribution and is a limitation, not an invisible implementation detail.

Hamiltonian MPC mean momentum RMS improves from approximately 0.1258 under local coverage to 0.0304 with mixed coverage and 0.0239 globally. Unrestricted-force MPC changes from 0.181 to 0.0328 and 0.0240. Under global unshifted coverage all six force families achieve 100% strict success for both tested controller types. The corresponding SAC comparison remains substantially worse: global momentum RMS is approximately 0.422 for force, 0.373 for odd force and 0.370 for Hamiltonian models, against 0.264 for the privileged true-model SAC reference. The Hamiltonian/odd gap is small relative to the controller optimization gap.

These results distinguish performance of a fitted model from performance of a finite-budget optimizer. They do not prove SAC asymptotic failure or a universal superiority of feedback/MPC. The known-force references are privileged and are not theoretical lower bounds for every finite sampled controller.

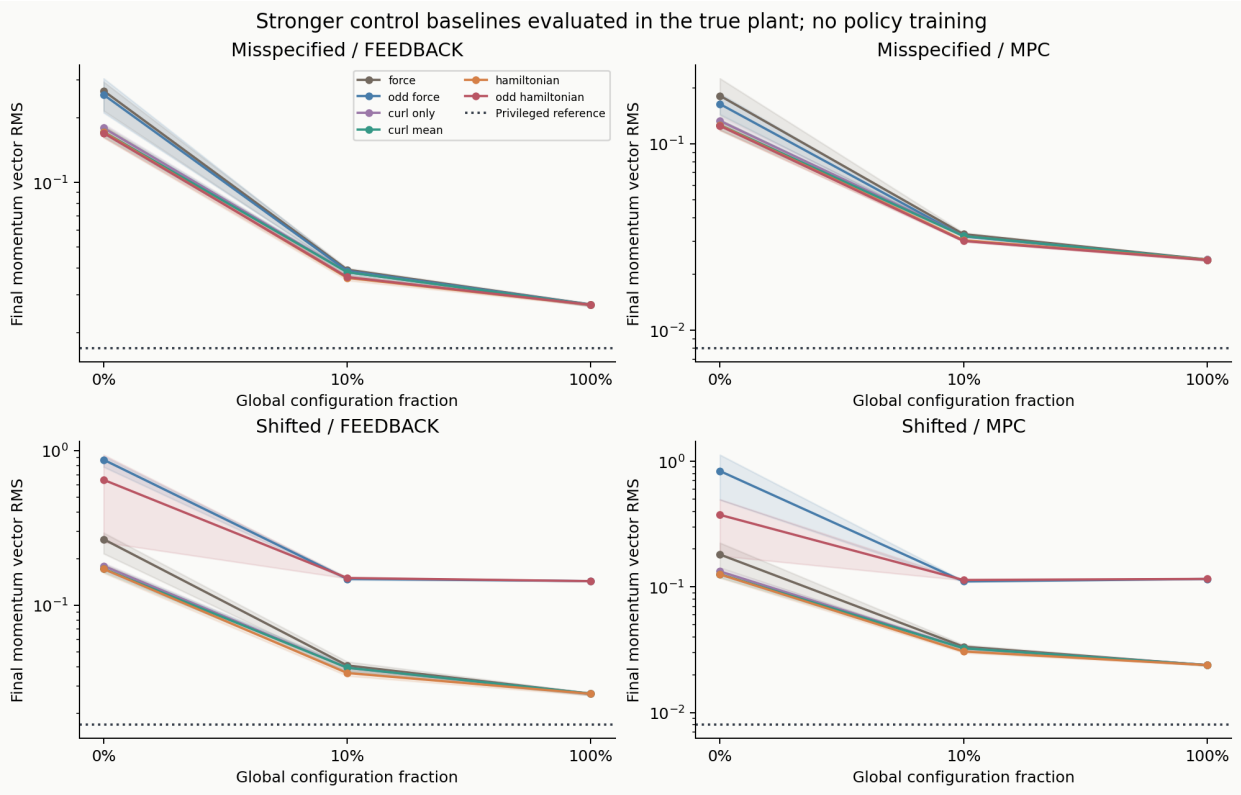


Figure 2: True-plant control with feedback and predictive control.

5.3 Frozen acquisition screen

The acquisition study starts with 128 noisy low-energy labels and adds 16, 64 or 256 selected labels from an unlabelled 1,024-configuration pool. It compares random sampling, pointwise uncertainty, global A-design, frozen-occupancy A-design and spectral task A-design. Five paired world seeds, two plant conditions, three force spaces and two target momenta produce 660 unique fitted models. There are 964 actual control evaluations; pilot aliases increase reported rows to 1,200 and are not additional replications. Across paths, 53,760 added-query calls use 8,001 distinct shared plant/world/pool-index label keys, plus 1,280 unique pilot labels.

The clipped controller runs for 20 time units at held interval 0.05. Sixty-four fresh uniform configuration phases start exactly at target momentum. The primary metric is momentum-vector RMS over the final five time units. This is tracking, not swing-up. The frozen practical gate requires at least 10% improvement over the best of four baselines and at least four of five paired-seed wins in every Hamiltonian B=64 cell, plus bounded degradation at other budgets.

Plant/task	Best baseline RMS	Spectral RMS	Degradation	Seed wins
canonical, forward	0.016906	0.017547	3.79%	2/5
canonical, swapped	0.015687	0.016543	5.46%	1/5
misspec., forward	0.034552	0.039674	14.83%	0/5
misspec., swapped	0.057841	0.059162	2.28%	2/5

The candidate fails every primary cell. Random is the best mean baseline in those cells, established after comparing all four baselines. The gate is descriptive, not a significance test, and 64 phases do not create 64 independent fitted-world seeds. The new-algorithm route is stopped; no rescue variants, additional candidate tuning or conditional MPC confirmations are introduced.

In the misspecified forward task, spectral acquisition reduces posterior tracking standard deviation from random's 0.00470 to 0.00411, but increases actual RMS from 0.03455 to 0.03967. The uncertainty criterion excludes

omitted-mode bias. This observed divergence is stronger evidence against the candidate's practical promise than a favorable covariance-only plot would be evidence for it.

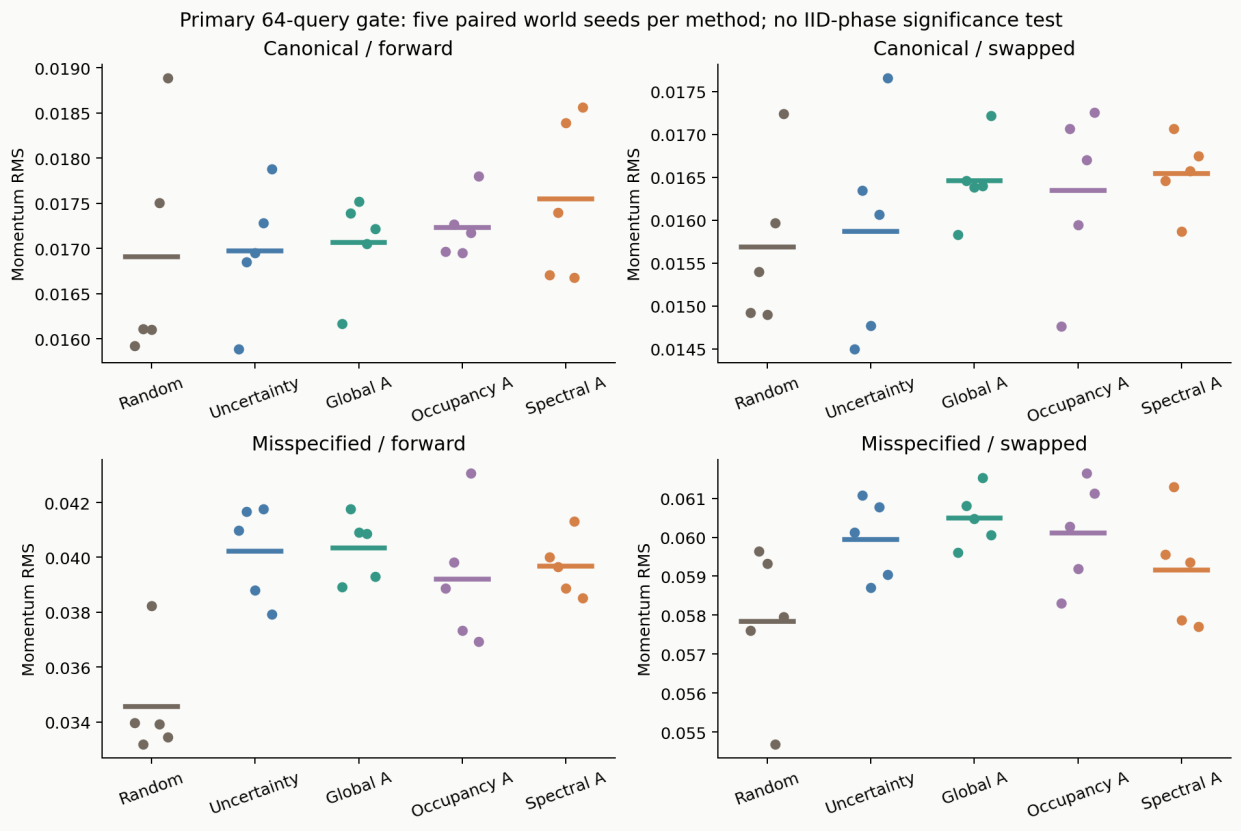


Figure 3: Frozen acquisition outcomes retain all five world seeds.

5.4 Independent validation and hold refinement

Coupled Duffing validation produces 45 fits. Mean global force RMS is:

Coverage	Force (20)	Odd force (12)	Hamiltonian (14)
local	0.50148	0.48487	0.40054
10% global	0.02146	0.01634	0.01755
global	0.00639	0.00511	0.00507

Coverage dominates the between-class difference; odd force slightly improves on the Hamiltonian mean under mixed coverage. This is independent-domain support for the coverage interpretation, with explicit dimension and fixed-ridge limitations. It is not another acquisition experiment.

The exact-force unsaturated action-hold diagnostic uses 32 shared phases over ten time units and evaluates the final five. Hold intervals 0.1, 0.05, 0.025 and 0.0125 give momentum RMS 0.035818, 0.017367, 0.008557 and 0.004248. The fitted log-log slope is 1.025. The observed refinement supports a first-order hold floor in that diagnostic, consistent with the finite-horizon expansion. It does not quantify every source of error in the clipped primary test.

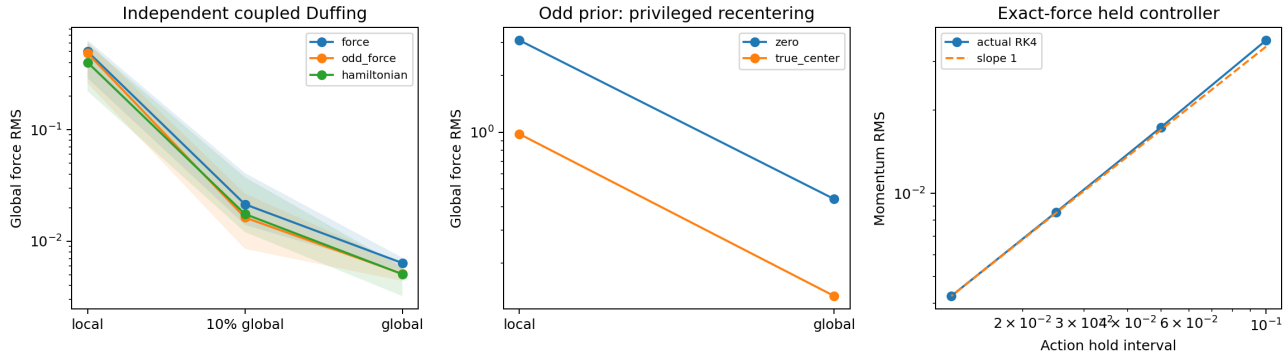


Figure 4: Final validation: independent dynamics, privileged recentering and exact-force action-hold refinement. Shaded Duffing bands show minimum/maximum across five seeds, not confidence intervals.

5.5 Simulation storyboards and interpretation

Figures below are extracted from the archived numerical animations, not generated illustrations. The same recorded case is used across every frame: misspecified periodic plant, Hamiltonian force head, world seed 20 and forward target. The control storyboard additionally fixes held-out initial phase zero and B=64. Its six panels show one trajectory under two acquisition rules at three times; they are not six independent experiments. The rods depict angular coordinates with separate pivots and an indicated coupling, not the geometry of a serial double pendulum.

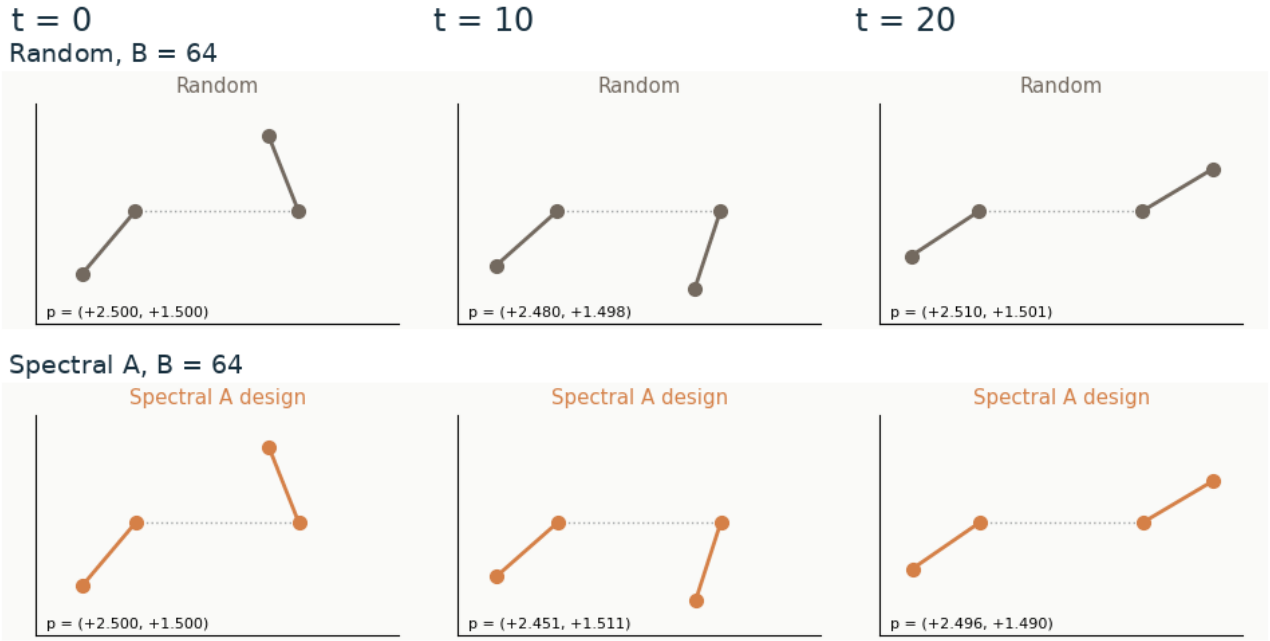


Figure 5: Exact control-animation frames at t=0, 10 and 20. Top: random acquisition; bottom: spectral A-design, each with 64 added labels and the same 128-label pilot. Momentum annotations are retained from the actual RK4 records; full tracking traces remain available in the GIF. A fixed illustrative trajectory does not determine the aggregate gate.

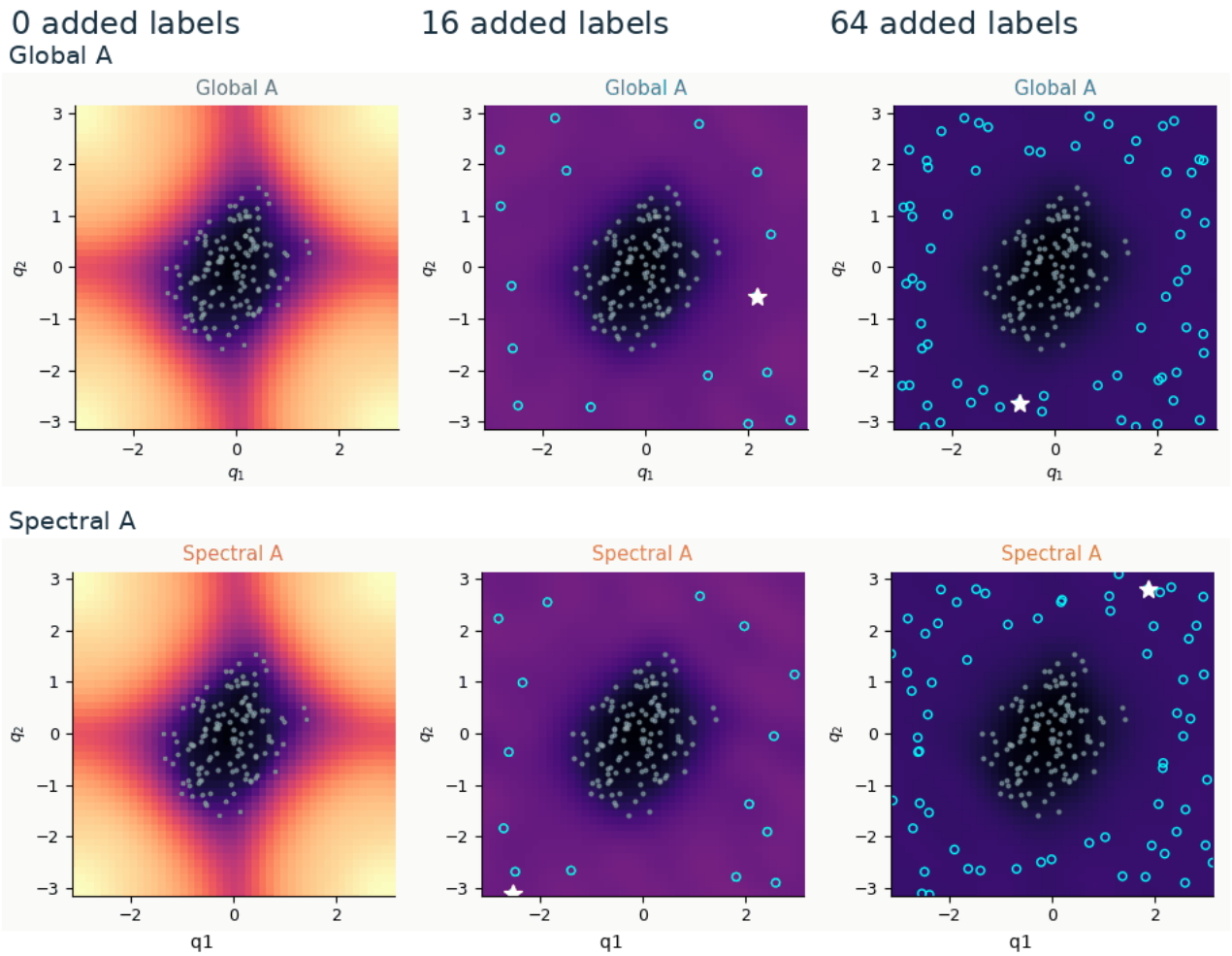


Figure 6: Sequential acquisition storyboard at 0, 16 and 64 added labels. Grey dots show the shared local pilot; cyan rings are queried configurations and stars mark the latest query. Top: global A-design; bottom: spectral A-design. Frames use the identical logarithmic variance scale shown in the following complete frame. Query count is not physical time, and posterior variance excludes approximation bias.

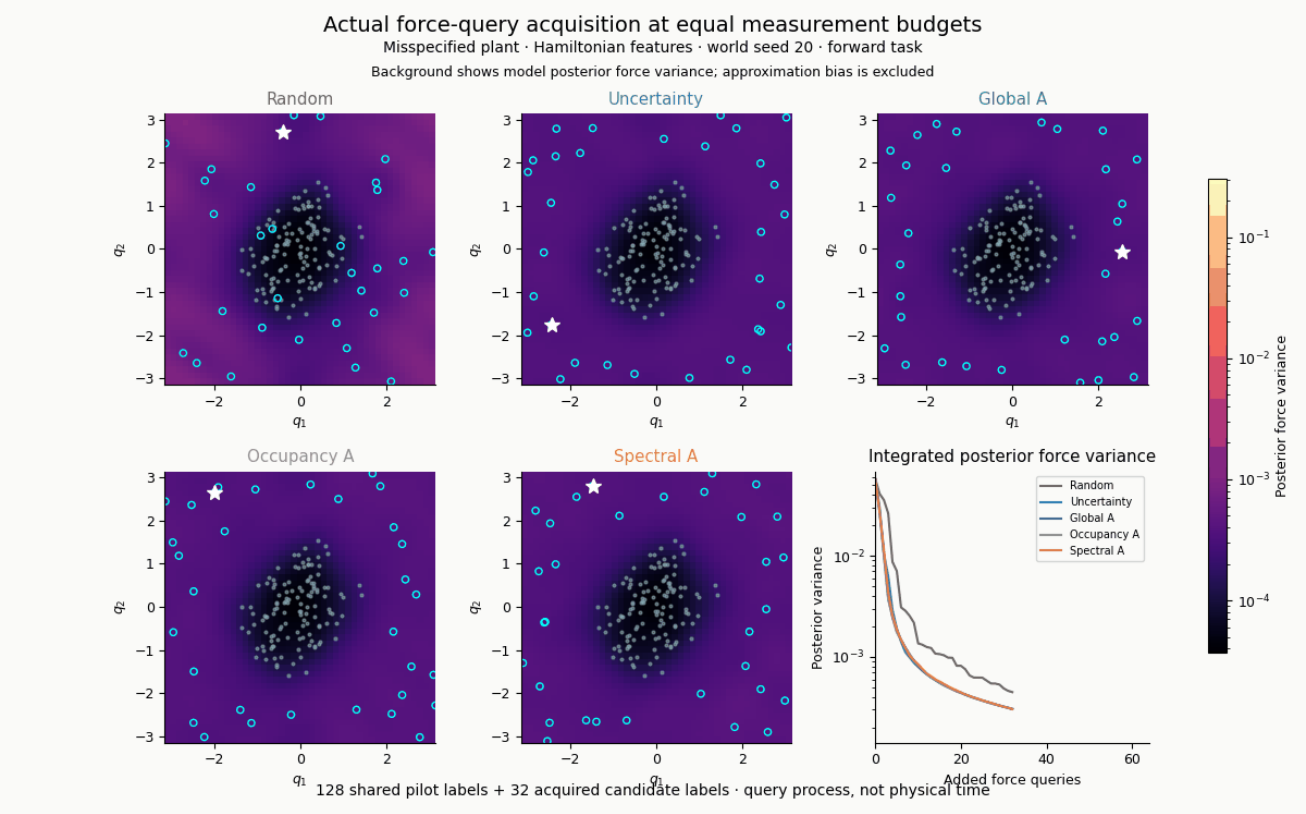


Figure 7: Complete acquisition-animation frame at 32 added labels, retaining the shared logarithmic colorbar, all five methods and the integrated-variance history. This frame complements the storyboard without introducing a new numerical evaluation.

The corresponding GIFs are `phase7/phase7_control_comparison.gif` and `phase7/phase7_acquisition.gif`. They play in the repository README and offline demo; the PDF supplies static multi-frame summaries. Extraction indices, crop bounds and source hashes are recorded in `paper/figures/animation_stills_provenance.json`.

6 Reproducibility and verification

The repository contains frozen protocols, raw JSON/CSV records, actual fitted checkpoints, code, PNG/SVG figures and real trajectory GIFs. Primary outcomes and posthoc diagnostics are labelled separately. Earlier failed optimization pilots remain available. No trained checkpoint is represented as an independently validated universal policy.

Regression normal equations, independent constrained-force fits, finite-difference controller gradients, Fourier Parseval identities and label-query audits check different failure modes. The final Duffing force/gradient check has maximum error 9.48×10^{-11}; an independent augmented-QR solution matches a Hamiltonian normal-equation fit within 2.10×10^{-15}. The largest final normal-equation residual is 4.18×10^{-15}. Acquisition verification checks selected labels only, confirms unique indices and protocol fingerprints, and reproduces the unchanged failed gate.

Full reproduction and a short demonstration are separated. The complete archive preserves all previous stages; a lightweight GitHub-ready bundle supplies source, protocols, figures, paper and numerical summaries without bulky replay/checkpoints. `ARTIFACT_INDEX.md` maps every output to its role. The final release records local validation and does not claim a remote GitHub publication or a DOI.

7 Discussion and limitations

The strongest supported conclusion is conditional: structural force restrictions can reduce local estimation variance, but coverage, coordinate correctness and controller optimization can dominate practical rankings. Hamiltonian consistency and conserved learned energy provide useful diagnostics, not identification guarantees. A task-aware posterior criterion can also be an inadequate proxy for tracking under approximation bias and sampled control.

There are important limits. Both periodic plants share mechanics; the translated probe is not a new physical environment. Duffing validation is estimation-only and uses a correctly specified polynomial potential. All observations are synthetic, all multidimensional comparators know mass and ports, and no contact-rich or dissipative robotic plant is tested. Trainable deep-feature models are not exhaustively compared; the strongest main results concern exact finite-dimensional force heads. Five acquisition seeds support a descriptive screen, not broad statistical generalization. We do not provide full BARL, DYNAMICOED or published kernel-method replications. The prior literature review is scoped and cannot certify absolute absence of prior art.

The action-hold analysis deliberately uses an unsaturated reference to isolate consistency. The main clipped controller can exhibit saturation and larger-model-error effects. Likewise, posterior covariance excludes approximation bias, and the fixed-ridge noise calculation excludes hyperparameter-selection randomness. These distinctions prevent overinterpreting the analytical identities.

The manuscript's originality lies in its executed, inspectable comparison and the specific combination of interventions and negative results. Its mathematics provides a rigorous synthesis of established tools plus project-specific consistency calculations. A novel general algorithm, a new general optimal-design theorem and general HNN superiority are not contributions of this work. Independent review may judge the empirical contribution appropriate for a workshop, reproducibility study or technical report rather than a main-conference methods paper.

8 Conclusion

Within the tested settings, Hamiltonian structure helps most when the force class is valid and observations are local. Broad coverage reduces that advantage, wrong coordinate-dependent symmetry introduces a removable floor, and finite-budget RL optimization can obscure improved world models. The failed acquisition screen and action-hold refinement show why task-aware uncertainty reduction is insufficient by itself. The released evidence supports a reproducible conditional account of Hamiltonian learning for control, with unsuccessful hypotheses retained as part of the research record.

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Mathematical appendix: detailed derivations

The following calculations expand the analysis behind the experiments. All norms on the torus use normalized Haar measure. Statements about exact continuous dynamics must be separated from statements about the integrators and held controls. Established identities are not claimed as new general results.

A. Canonical structure, work balance and non-identifiability

For $z = (q, p)$ and $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$, a controlled Hamiltonian vector field is

$$\dot{z} = J\nabla H(z) + Gu, \qquad G = \begin{pmatrix} 0 \\ I \end{pmatrix}.$$

Since $J^\top = -J$,

$$\frac{dH}{dt} = \nabla H^\top J \nabla H + \nabla H^\top G u = p^\top u,$$

$$H(z(T)) - H(z(0)) = \int_0^T p(t)^\top u(t)\, dt.$$

The experiment therefore measures a work-balance residual, not raw energy drift during actuation. For a learned force $\widehat{F}$ used without torque, the true-energy derivative is instead

$$\frac{dH(q, \widehat{p})}{dt} = \widehat{p}^\top [\widehat{F}(q) - F(q)].$$

This term can be nonzero even if the learned Hamiltonian is conserved exactly. To obtain a conditional quantitative bound, assume $\|\widehat{F} - F\| \le \delta$ and $\|\widehat{p}\| \le P$ along the entire learned trajectory. Then

$$|H(\widehat{z}(T)) - H(\widehat{z}(0))| \le TP\delta.$$

A small training-domain force error does not establish either bound along an unseen trajectory.

For explicit non-identifiability, let U contain all observed configurations and choose a smooth bump b supported in an open region disjoint from U. Define

$$\widetilde{V}(q) = V(q) + ab(q), \qquad \widetilde{F}(q) = F(q) - a\nabla b(q).$$

On U, all force labels and all derivatives of the perturbation vanish. Both vector fields are Hamiltonian, yet outside U their difference can be scaled by arbitrary finite a. This construction assumes an unrestricted smooth potential class; a fixed finite analytic feature class may instead be identifiable from sufficiently informative samples. Poor conditioning can nevertheless make finite noisy extrapolation inaccurate. The smooth bump argument must not be misrepresented as exact finite-Fourier non-identifiability.

B. Fourier curl, periods and gradient-force equivalence

Write $F = c + \sum_k (A_k \cos k^\top q + B_k \sin k^\top q)$ and let $k^\perp = (-k_2, k_1)$. Then

$$\partial_{q_1} F_2 - \partial_{q_2} F_1 = \sum_k [-(k^\perp \cdot A_k)\sin(k^\top q) + (k^\perp \cdot B_k)\cos(k^\top q)].$$

Orthogonality implies curl vanishes exactly when A_k and B_k are parallel to k for every mode. This does not remove c. For each coordinate-cycle at a fixed transverse coordinate, all nonconstant mode contributions integrate to zero once the curl-free parallel condition holds, and the period is

$$\oint_{\gamma_j} F^\top dq = 2\pi c_j.$$

Consequently, a finite periodic curl-free force admits a periodic scalar potential precisely when $c = 0$. If $A_k = a_k k$ and $B_k = b_k k$, one potential satisfying $F = -\nabla V$ is

$$V(q) = \sum_k [b_k \cos(k^\top q) - a_k \sin(k^\top q)] + C.$$

Thus a Hamiltonian feature head is exactly a constrained-force regressor under matched objectives. Parameterizing the gradient through a scalar does not introduce a different statistical estimator if the constrained subspace, regularizer and labels are the same. The project verifies this equivalence numerically; differences from unconstrained fitting arise from the hypothesis class and objective.

Parseval gives

$$\|F\|_2^2 = \|c\|^2 + \frac{1}{2}\sum_k (\|A_k\|^2 + \|B_k\|^2),$$

$$\|\operatorname{curl} F\|_2^2 = \frac{1}{2}\sum_k [(k^\perp \cdot A_k)^2 + (k^\perp \cdot B_k)^2].$$

These exact population quadratic forms generate the force-norm and curl penalties; numerical quadrature is unnecessary for those penalties.

C. Translation and the parity projection floor

For $\theta_k = k^\top s$, translation $q \mapsto q - s$ rotates coefficients:

$$\begin{pmatrix} A_k' \\ B_k' \end{pmatrix} = \begin{pmatrix} \cos\theta_k & -\sin\theta_k \\ \sin\theta_k & \cos\theta_k \end{pmatrix} \begin{pmatrix} A_k \\ B_k \end{pmatrix}.$$

The rotation acts separately on each force component and preserves Parseval norm. Parallel coefficients remain parallel, hence Hamiltonian closure. Sine-only force is not closed under this action.

A true included force mode $-ak\sin(k^\top(q-s))$ expands as

$$-ak\cos\theta_k \sin(k^\top q) + ak\sin\theta_k \cos(k^\top q).$$

Orthogonal projection onto the zero-centered odd space drops the second term. Its squared error is $a^2\|k\|^2\sin^2\theta_k/2$. Summing included modes and the orthogonal missing frequency yields the approximation-floor expression in the main text. This is a population-class minimum. Empirical test estimates may fluctuate below it, and ridge fits from local data need not approach it.

D. Exact fixed-design bias-variance decomposition

Stack componentwise force observations as $y = X\theta_* + r + \epsilon$, where r is a deterministic class-misspecification residual at the observed locations. Let X denote the subspace design, $D = X^\top X/N$, regularization Λ and $A = D + \Lambda$. The fitted coefficients are

$$\widehat{\theta} = A^{-1}X^\top y/N,$$

$$\overline{\theta} := \mathbb{E}_\epsilon \widehat{\theta} = \theta_* - A^{-1}\Lambda\theta_* + A^{-1}X^\top r/N.$$

The first bias term is shrinkage; the second is observed-design aliasing of omitted dynamics. Orthogonality under a global population measure does not force $X^\top r = 0$ under a local or finite design.

Let R be the test-domain coefficient Gram matrix and r_{test} the residual of the chosen reference representation. The total expected test error decomposes as

$$\mathbb{E}_\epsilon \|\widehat{F} - F_*\|_{\text{test}}^2 = \|F_{\bar{\theta}} - F_*\|_{\text{test}}^2 + \frac{\sigma^2}{N} \operatorname{tr}(RA^{-1}DA^{-1}).$$

The cross term vanishes because the centered estimator has zero mean. This formula accommodates approximation and design aliasing through the first term; the posterior covariance $\sigma^2 A^{-1}$ is generally different from frequentist noise covariance $\sigma^2 A^{-1}DA^{-1}/N$ in the averaged-loss convention. The two expressions use different assumptions and precision scaling and must not be interchanged.

For a correctly specified, unregularized full-rank design, $A = D$ and variance is $\sigma^2 \operatorname{tr}(RD^{-1})/N$. If D is close to R, the leading term is $\sigma^2 d/N$. With fixed ridge and population whitening, diagonalizing D gives the sum in the main text. With validation-selected ridge, conditioning on that selected value does not generally restore an unconditional fixed-ridge identity because selection depends on noise.

E. Full periodic filter calculation and finite-time correction

For one vector-valued mode $f(t) = A\cos(\theta+\omega t) + B\sin(\theta+\omega t)$, seek $e_s = C\cos(\theta+\omega t) + D\sin(\theta+\omega t)$. Matching coefficients in $\dot{e}_s + \kappa e_s = f$ gives

$$\kappa C + \omega D = A, \qquad -\omega C + \kappa D = B,$$

$$C = \frac{\kappa A - \omega B}{\kappa^2 + \omega^2}, \qquad D = \frac{\omega A + \kappa B}{\kappa^2 + \omega^2}.$$

The cross terms cancel in the response norm:

$$\|C\|^2 + \|D\|^2 = \frac{\|A\|^2 + \|B\|^2}{\kappa^2 + \omega^2}.$$

Averaging over uniform torus phase makes different frequency representatives orthogonal and gives the steady formula. If $e(0) = 0$, the finite-time solution is

$$e_1(t) = e_s(q_0 + p_* t) - e^{-\kappa t} e_s(q_0).$$

This includes the DC contribution $c(1 - e^{-\kappa t})/\kappa$. The steady expression is therefore not exactly the finite-time zero-initial-error metric. For each mode, uniform-phase squared error becomes

$$\frac{\|A_k\|^2 + \|B_k\|^2}{2(\kappa^2 + \omega_k^2)}[1 + e^{-2\kappa t} - 2e^{-\kappa t}\cos(\omega_k t)].$$

The DC squared term is $\|c\|^2(1 - e^{-\kappa t})^2/\kappa^2$. On the final test window with $\kappa = 2$, this linear-model transient is negligible, but nonlinear phase drift, saturation and held actions are not thereby negligible. At resonance $\omega_k = 0$, the formula remains well-defined; it concerns an ensemble of initial phases, not temporal decorrelation along one resonant trajectory.

F. Weighted design, Woodbury and invariant information

Let $L = T(T^\top RT)^{-1/2}$ and $B = \Phi L$. Since $\int \Phi^\top \Phi \, dq = R$,

$$\int B(q)^\top B(q) \, dq = L^\top RL = I.$$

The scalar Gaussian prior has precision $\rho I/\sigma^2$. A two-component observation adds $B^\top B/\sigma^2$. Setting $A = \rho I + \sum B_i^\top B_i$, Woodbury gives

$$A_{\text{new}}^{-1} = A^{-1} - A^{-1}B^\top (I_2 + BA^{-1}B^\top)^{-1} BA^{-1}.$$

For a positive-semidefinite task metric W, the exact candidate gain is

$$g_W(q) = \sigma^2 \operatorname{tr}\left[WA^{-1}B^\top (I_2 + BA^{-1}B^\top)^{-1} BA^{-1}\right] \geq 0.$$

Nonnegativity follows because the covariance reduction is positive semidefinite. A positive scalar rescaling of W changes all gains by that scalar and cannot change the selected maximizer. These gains depend on locations and covariance, not observed force values, for this fixed Gaussian linear model.

For completeness, convexity of $\Psi(M) = \operatorname{tr}[W(\rho I + nM)^{-1}]$ can be checked along a symmetric direction H. With $S = (\rho I + nM)^{-1}$,

$$\frac{d^2}{dt^2}\Psi(M + tH)\bigg|_{t=0} = 2n^2 \operatorname{tr}(WSHSHS) \geq 0.$$

Indeed $SHSHS = S^{1/2}(S^{1/2}HS^{1/2})^2 S^{1/2}$ is positive semidefinite. Translation invariance then permits Jensen averaging as in the main proof. A fixed pilot matrix A_0 that does not commute with the translations replaces ρI and invalidates the invariance step. Consequently the idealization is not a theorem that the observed random design must win.

G. Predictive-control adjoint and SAC objectives

The MPC model uses kick-drift updates,

$$p_{t+1} = p_t + h[\widehat{F}(q_t) + u_t], \qquad q_{t+1} = q_t + hp_{t+1}.$$

Writing $J_t = D\widehat{F}(q_t)$, the state and input Jacobians are

$$A_t = \begin{pmatrix} I + h^2 J_t & hI \\ hJ_t & I \end{pmatrix}, \qquad B_t = \begin{pmatrix} h^2 I \\ hI \end{pmatrix}.$$

For an objective $C = \sum_{t=0}^{H-1} \ell(z_{t+1}, u_t) + \phi(z_H)$, initialize $\lambda_H = \nabla\phi(z_H)$. A backward step has

$$g_{u_t} = \partial_u \ell + B_t^\top [\lambda_{t+1} + \nabla_z \ell], \qquad \lambda_t = A_t^\top [\lambda_{t+1} + \nabla_z \ell].$$

Here $\ell = h(\|p_{t+1} - p_*\|^2 + 0.01\|u_t\|^2)$ and $\phi = 0.5\|p_H - p_*\|^2$. Projected-gradient backtracking guarantees only nonincrease of this fitted-model objective. It supplies no true-plant monotonicity or global optimum.

The actual coupled SAC implementation observes

$$o(z) = (\cos q_1, \cos q_2, \sin q_1, \sin q_2, p_1/3, p_2/3),$$

and uses squashed Gaussian torque $u = 2\tanh a$. Its change-of-variables density is

$$\log \pi(u \mid o) = \sum_j \left[\log \mathcal{N}(a_j; \mu_j, \sigma_j^2) - \log 2 - \log(1 - \tanh^2 a_j)\right],$$

with a small numerical logarithm safeguard in code. Twin critics use target

$$y = r + \gamma[\min_i \overline{Q}_i(o', u') - \alpha \log \pi(u' \mid o')],$$

$$\mathcal{L}_Q = \sum_{i=1}^{2} \mathbb{E}(Q_i(o,u) - y)^2, \qquad \mathcal{L}_\pi = \mathbb{E}[\alpha \log \pi(u \mid o) - \min_i Q_i(o,u)].$$

In the coupled implementation, $\gamma = 0.99$, fixed $\alpha = 0.01$, Adam learning rate 3×10^{-4} and target interpolation rate 0.005. Reward integrates $-\|p - p_*\|^2 - 0.01\|u\|^2$ by substep trapezoidal quadrature. Replay training uses a continuing-task target without a terminal multiplier; finite evaluation episodes and recorded training resets are therefore distinct from a fully specified episodic Bellman model. These details limit interpretation of finite-budget policy rankings. The formulas document the implementation; they are standard SAC objectives, not a new RL method.

H. Detailed action-hold consistency and missing-mode tracking

At a grid state (q_n, p_n) with $e_n = p_n - p_*$, exact-force cancellation yields $\dot{p}(t_n) = -\kappa e_n$ while the held torque has zero derivative inside the interval. Hence

$$p_{n+1} = p_n - \kappa h e_n + \frac{h^2}{2} DF(q_n) p_n + O(h^3).$$

A finite-horizon boundedness induction and discrete Gronwall give $\sup_n \|e_n\| \le C_T h$. Integrating $\dot{q} = p_* + e$ gives $q_n = q_0 + p_* t_n + O(h)$. Set $r_n = e_n/h$ and use bounded second derivatives to replace $DF(q_n)p_n$:

$$r_{n+1} = r_n + h\left[-\kappa r_n + \frac{1}{2} DF(q_0 + p_* t_n) p_*\right] + O(h^2).$$

Comparing to forward Euler for the response ODE proves $r_n = \eta(t_n) + O(h)$. For C^2 periodic forces on a compact torus and finite T, the derivative assumptions hold; an unbounded configuration-space system requires a separate bounded-trajectory neighborhood. Plant integration error must be controlled separately. The diagnostic fixes RK4 substeps at 0.0025 while varying action-hold interval, so control sampling is much coarser than integration.

For the periodic missing potential $\varepsilon[1 - \cos(k_*^\top q)]$, its force amplitude is $-\varepsilon k_* \sin(k_*^\top q)$. The ideal first-order squared tracking contribution is

$$\frac{\varepsilon^2 \|k_*\|^2}{2[\kappa^2 + (k_*^\top p_*)^2]}.$$

With $k_* = (3, -1)$ and the two targets, $\omega_* = 6$ or 2; at $\kappa = 2$ and $\varepsilon = 0.06$, RMS contributions are 0.021213 and 0.047434. These are ideal linear response values, not exact nonlinear tracking lower bounds. The Fourier force approximation floor, by contrast, is an exact class-level statement.